HW #6

$\boxed{2.32}$

$$\begin{cases} C\sin(la) + D\cos(la) = F e^{ika} & \longrightarrow (A) \\ C\cos la - D\sin(la) = \frac{ikF}{l} e^{ika} & \longrightarrow (B) \end{cases}$$

multiply Eqn (A) & (B)
$\downarrow \sin(la)$ $\quad \downarrow \cos(la)$

$$\Rightarrow \begin{cases} C\sin^2(la) + D\sin(la)\cos(la) = F\sin la\, e^{ika} \\ C\cos^2(la) - D\sin(la)\cos(la) = \frac{ikF}{l}\cos la\, e^{ika} \end{cases}$$

Add the two Eqns

$$\Rightarrow C = e^{ika} F\left[\sin(la) + \frac{ik}{l}\cos(la)\right] \longrightarrow (1)$$

Multiply Eqns (A) by $\cos la$
(B) by $\sin la$

$$\begin{cases} C\cos(la)\sin(la) + D\cos^2(la) = F\cos(la)\, e^{ika} \\ C\sin la\cos(la) - D\sin^2 la = \frac{ikF}{l}\sin(la)\, e^{ika} \end{cases}$$

Subtract the two Eqns

$$D = Fe^{ika}\left[\cos(la) - \frac{ik}{l}\sin(la)\right] \longrightarrow (2)$$

We are solving for (A) & (B)

$$\begin{cases} Ae^{-ika} + Be^{ika} = -C\sin(la) + D\cos(la) \\ Ae^{-ika} - Be^{ika} = \frac{l}{ik}\left[C\cos(la) + D\sin(la)\right] \end{cases}$$

$$(3) \longleftrightarrow A\left(2e^{-ika}\right) = C\left[-\sin la + \frac{l\cos la}{ik}\right] + D\left[\cos(la) + \frac{l\sin(la)}{ik}\right]$$

$$(4) \longleftarrow B\left(2e^{ika}\right) = C\left[-\sin la - \frac{l\cos la}{ik}\right] + D\left[\cos la - \frac{l\sin la}{ik}\right]$$

~~Plug the C & D values obtained~~

Plug 1 & 2 in 3 & 4 to get:

$$B = i\,\frac{\sin(2la)}{2kl}\,(l^2-k^2)\,F$$

$$F = \frac{e^{-2ika}\,A}{\cos(2la) - i\,\frac{(k^2+l^2)}{2kl}\,\sin(2la)}$$

$$T = \frac{|F|^2}{|A|^2} \Rightarrow T^{-1} = \left|\frac{A}{F}\right|^2 = \cos^2(2la) + \left(\frac{k^2+l^2}{2kl}\right)^2 \sin^2(2la)$$

$$= 1 + \sin^2(2la)\left[\left(\frac{k^2+l^2}{2kl}\right)^2 - 1\right]$$

$$\cancel{\Rightarrow (k^2+l^2)^2 - 4k^2l^2 = (k+l)^2(k-l)^2}$$

$$\cancel{= \left(\frac{2mE}{\hbar} + \frac{2m(E+V_0)}{\hbar}\right)^2 -}$$

$$(k^2+l^2) - 4k^2l^2 = (k^2+l^2-2kl)(k^2+l^2+2kl)$$

$$= (k-l)^2(k+l)^2$$

$$= (k^2-l^2)^2$$

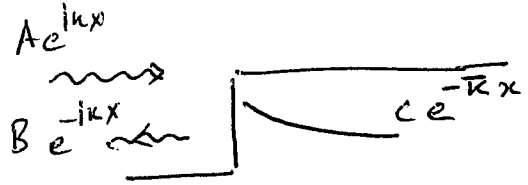
$$= \left(\frac{2mE}{\hbar} - \frac{2m(E+V_0)}{\hbar}\right)^2$$

$$= \left(\frac{2mV_0}{\hbar}\right)^2$$

$$(2kl)^2 = 4\left(\frac{2mE}{\hbar}\right)\left(\frac{2m(E+V_0)}{\hbar}\right) = 4\left(\frac{2m}{\hbar}\right)^2 E\,(E+V_0)$$

$$T^{-1} = 1 + \frac{V_0^2}{4E(E+V_0)}\sin^2\left[\frac{2a\sqrt{2m(E+V_0)}}{\hbar}\right]$$

$\boxed{2.34}$ (a)



$$k = \sqrt{\frac{2mE}{\hbar^2}}$$

$$\bar{k} = \sqrt{\frac{2m(V_0 - E)}{\hbar^2}}$$

$\boxed{E < V_0}$

$$R = \text{Reflection Coefficient} \equiv \left| \frac{J_{Reflected}}{J_{Incident}} \right|$$

$$T \equiv \text{transmission Coefficient} \equiv \left| \frac{J_{Transmitted}}{J_{Incident}} \right|$$

We know that $\rightarrow$ $R + T = 1$

$$\Rightarrow J_{Reflected} = J_{Incident} - J_{transmitted}$$

$$J_{transmitted} = \frac{-i\hbar}{2m}\left[\psi^* \frac{\partial \psi}{\partial x} - \psi \frac{\partial \psi^*}{\partial x}\right]$$

where $\psi \equiv c e^{-\bar{k}x}$

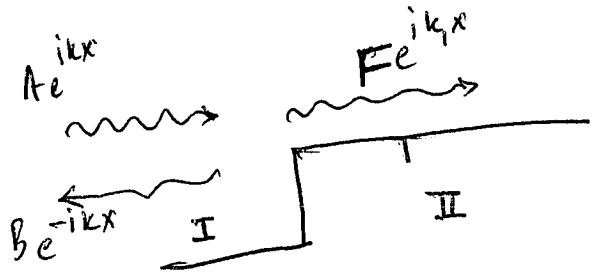
plugging it in we find

$$J_{transmitted} = 0$$

$$\Rightarrow J_{reflected} = J_{incident} \Rightarrow T = 0$$

$$\Rightarrow R = 1$$

(b) $E > V_0$



$$k = \sqrt{\frac{2mE}{\hbar}}$$

$$k_1 = \sqrt{\frac{2m(E-V_0)}{\hbar}}$$

$$J_{Incident} = \frac{-i\hbar}{m}\left[\psi^* \frac{\partial \psi}{\partial x} - \psi \frac{\partial \psi^*}{\partial x}\right] \text{ ; where } \psi = Ae^{ikx}$$

$$= \frac{\hbar k}{m}|A|^2$$

$$J_{reflected} \xrightarrow[\text{use } \psi = Be^{-ikx}]{} -\frac{\hbar k}{m}|B|^2$$

$$J_{transmitted} \xrightarrow[\text{use } \psi = Ce^{ik_1x}]{} \frac{\hbar k_1}{m}|F|^2$$

From BC $\rightarrow$

$$\left.\begin{aligned} \psi_I(x) &= Ae^{ikx} + Be^{-ikx} \\ \psi_{II}(x) &= Fe^{ik_1x} \end{aligned}\right\} \quad \begin{aligned} &\psi_I(0) = \psi_{II}(0) \\ &\left.\frac{\partial \psi}{\partial x}\right|_I = \left.\frac{\partial \psi}{\partial x}\right|_{II} \end{aligned}$$

From the continuity of the wave function $\leftarrow$ $A + B = F$

From the differentiability $\leftarrow$ $ik[A - B] = ik_1 F$

$$A + B = F$$

$$A - B = \left(\frac{k_1}{k}\right) F$$

$$\Rightarrow \quad 2A = F\left[1 + \frac{k_1}{k}\right] \Rightarrow \frac{A}{F} = \frac{\left(1 + \frac{k_1}{k}\right)}{2}$$

and

$$\left.\begin{aligned} A + B &= F \\ \frac{k}{k_1}(A - B) &= F \end{aligned}\right\} \quad (A+B) - \frac{k}{k_1}(A - B) = 0$$

$$A\left[1 - \frac{k}{k_1}\right] + B\left[1 + \frac{k}{k_1}\right] = 0$$

$$R = \left|\frac{J_{reflected}}{J_{Incident}}\right| = \frac{\frac{\hbar k}{m}|B|^2}{\frac{\hbar k}{m}|A|^2} = \frac{\left(\frac{k_1}{k} - 1\right)^2}{\left(1 + \frac{k}{k_1}\right)^2} \qquad \frac{A}{B} = \frac{\left(1 + \frac{k}{k_1}\right)}{\left(\frac{k_1}{k} - 1\right)}$$

use to obtain reflection coefficient

(c) $$T = \frac{J_{Transmitted}}{J_{INCIDENT}} = \frac{\frac{\hbar k_1}{m}|F|^2}{\frac{\hbar k}{m}|A|^2} = \frac{k_1}{k}\left[\frac{4}{\left(1 + \frac{k_1}{k}\right)^2}\right]$$

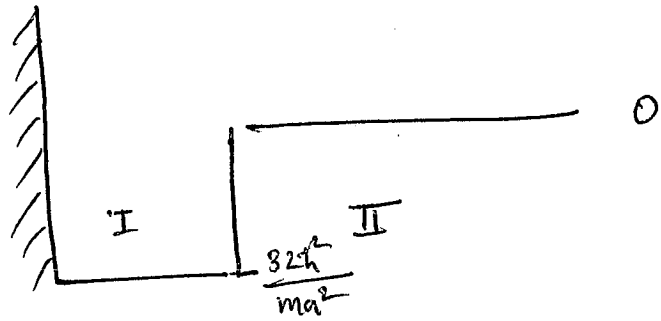
$$= \sqrt{\frac{E - V_0}{E}}\left[\frac{4}{\left(1 + \frac{k_1}{k}\right)^2}\right]$$

d $\quad R + T$

$$= \frac{\left(k_1/k - 1\right)^2}{\left(1 + \frac{k}{k_1}\right)^2} + \frac{4\left(\frac{k_1}{k}\right)}{\left(1 + \frac{k}{k_1}\right)^2}$$

$$= \frac{\left(1 + \frac{k_1}{k_1}\right)^2}{\left(1 + \frac{k}{k_1}\right)^2} = 1$$

2.40



$$V(x) = \begin{cases} \infty & x < 0 \\ -32\hbar^2/ma^2 & 0 \le x \le a \\ 0 & x > 0 \end{cases}$$

Region I

$$-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\psi + V\psi = \varepsilon\psi$$

since it is a bound state

$$\boxed{\varepsilon < 0}$$

hence $\rightarrow \varepsilon = -|\varepsilon|$

$$\Rightarrow -\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\psi = (-V - |\varepsilon|)\psi = \left[\frac{32\hbar^2}{ma^2} - |E|\right]\psi$$

Note also $\rightarrow$

$$-\frac{32\hbar^2}{ma^2} < -|\varepsilon| < 0$$

$$\Rightarrow \frac{32\hbar^2}{ma^2} - |E| > 0$$

hence oscillatory soln: with

$$k = \sqrt{\frac{2m}{\hbar^2}\left(\frac{32\hbar^2}{ma^2} - |E|\right)}$$

$$\psi_I(x) = A \sin kx + B \cos kx$$

$$\psi_I(0) = 0 \Rightarrow B = 0 \Rightarrow \psi_I(x) = A \sin kx$$

Boundry Condition

Region II → $V = 0$

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + \cancel{V}^{\,0}\psi = \varepsilon\psi$$

$$\Rightarrow -\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = -|\varepsilon|\psi$$

$\Rightarrow$ Exponential solution → $\bar{k} = \sqrt{\frac{2m|\varepsilon|}{\hbar^2}}$

$$\psi_{II}(x) = C e^{-\bar{k}x} + D e^{+\bar{k}x}$$

From Boundry Condition $\psi_{II}(\infty) = 0$

$$\Rightarrow D = 0$$

thus $\psi_{II}(x) = C e^{-\bar{k}x}$

Boundry Condition at $x=a$

$$\psi_I(a) = \psi_{II}(a) \Rightarrow A\sin ka = c\,e^{-\bar{k}a} \rightarrow (A)$$

$$\left.\frac{\partial \psi_I}{\partial x}\right|_{x=a} = \left.\frac{\partial \psi_{II}}{\partial x}\right|_{x=a} \Rightarrow Ak\cos ka = -c\bar{k}e^{-\bar{k}a} \rightarrow (B)$$

From (A) and (B)

$$\tan ka = -\frac{k}{\bar{k}}$$

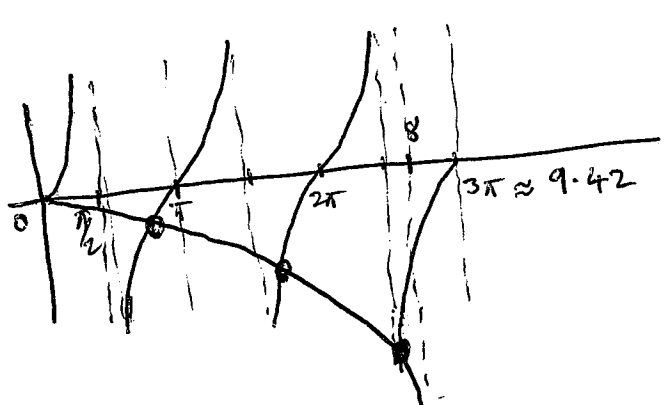
put $ka = z$

$$\Rightarrow \bar{k}a = a\sqrt{\frac{2m|E|}{\hbar^2}} = a\sqrt{\frac{2m}{\hbar^2}\left(\frac{32\hbar^2}{ma^2}\right) - \frac{k^2}{}}$$

$$= \sqrt{64 - z^2}$$

thus $\rightarrow$ $\tan ka = -\dfrac{ka}{\bar{k}a}$

$$\tan z = -\frac{z}{\sqrt{64 - z^2}}$$



$\Rightarrow$ Intersects at 3-places

$\Rightarrow$ 3-Bound State

$\boxed{2.23}$

$$\int_c^d f(x)\,\delta(x-a)\,dx = \begin{cases} f(a) & \text{if } c<a<d \\ 0 & \text{if } \begin{cases} a>d \\ \text{or } a<c \end{cases} \end{cases}$$

(a) $$\int_{-3}^{1} (x^3-3x^2+2x-1)\,\underbrace{\delta(x+2)}_{\delta(x-\underbrace{(-2)}_{a=-2})}\,dx$$

$a=-2$ lies in the interval $\rightarrow$ $-3<-2<1$

$$\Rightarrow \int_{-3}^{1} f(x)\,\delta(x+2)\,dx = f(-2) = -25$$

(b) $$\int_0^{\infty} [\cos(3x)+2]\,\delta(x-\pi)\,dx = (\cos 3\pi + 2) = 1$$

(c) $$\int_{-1}^{1} \exp[|x|+3]\,\delta(x-2)\,dx = 0$$

$a=2$ lies outside the region of integration!